Software Engineering Project May-2026

Milestone 1 Deliverables for ZenStudy

Submitted By

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Team Code : Team-051



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Community Services Platform for Library Management

Project Details

Software Name: ZenStudy

Department Context: Library Management System

System Type: Small Business Logistics & Community Services Platform

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Team Details

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PROBLEM STATEMENT

Study libraries and reading rooms are increasingly popular among students seeking quiet and dedicated spaces for learning. However, many study libraries continue to rely on manual or semi-digital processes to manage memberships, seat bookings, attendance, payments, and daily operations. These traditional methods often lead to inefficiencies, operational delays, and poor resource utilization.

Students frequently face challenges such as the lack of real-time information on seat availability, difficulty in reserving study spaces, limited visibility into their bookings and membership status, and inconvenience when requesting booking extensions or resolving issues. Manual check-in procedures and the absence of a centralized platform also result in longer waiting times and reduced user satisfaction.

From an operational perspective, library staff must manually manage walk-in admissions, verify bookings, process offline payments, allocate seats, track attendance, handle no-shows, and resolve complaints. These repetitive administrative tasks increase the likelihood of human error and make it difficult to efficiently utilize available study spaces.

Additionally, library administrators often lack comprehensive insights into occupancy trends, peak usage hours, membership statistics, and revenue, limiting their ability to make informed decisions regarding resource allocation and policy improvements.

Therefore, there is a need for a centralized Study Library Management System that streamlines membership management, seat booking, attendance tracking, payment recording, and administrative operations while providing real-time information to students, supporting efficient daily operations for staff, and enabling data-driven decision-making for administrators. Such a system would improve operational efficiency, optimize space utilization, and enhance the overall study library experience for all stakeholders.

IDENTIFICATION OF USERS

1. Primary Users:

These are the core operational users of the Study Library Management System. The platform's daily workflows, including membership verification, seat reservations, check-ins, payments and occupancy management, are primarily designed around their interactions. Their continuous use ensures the smooth functioning of the study library.

1.1 Library Staff: Library Staff are responsible for the day-to-day operations of the study library. Their responsibilities include:

- Registering new students and verifying memberships.
- Viewing daily bookings and expected arrivals.
- Verifying student bookings and performing check-ins using the student's unique check-in code.
- Handling same-day walk-in admissions by allocating available seats.
- Recording payments (cash, UPI, card, etc.).
- Marking no-shows and releasing unoccupied reserved seats.
- Managing bookings by approving or cancelling reservations when necessary.
- Handling student complaints and providing operational support.

The system must provide an efficient operational dashboard that enables staff to manage bookings, occupancy, payments, and student services with minimal manual effort.

1.2 Student: Students are the primary beneficiaries of the study library services and interact with the platform to access study facilities. Their responsibilities include:

- Logging in to access their account
- Viewing membership details and validity.
- Checking real-time seat and study room availability and payment history
- Booking study seats or rooms in advance.
- Modifying or cancelling bookings within the permitted period.
- Presenting their unique check-in code to library staff upon arrival.
- Checking out after completing a study session.
- Raising complaints or support requests when necessary.

The system must provide students with a simple and convenient platform for reserving study spaces, tracking their activity, and communicating with library staff.

2. Secondary Users:

These users oversee the overall functioning of the study library and are responsible for administrative decisions rather than daily operational activities.

Admin: Admin supervises the entire study library management system and ensures that policies and configurations are maintained effectively. Responsibilities include:

- Managing library staff accounts.
- Approving and managing student memberships.
- Configuring operating hours, booking limits and membership plans.
- Viewing reports and statistics related to occupancy, peak hours, memberships and revenue.
- Managing announcements and notifications for users.

The system must provide comprehensive administrative controls, configurable system settings, and analytical dashboards to support informed decision-making and efficient management.

3. Tertiary Users:

The proposed Study Library Management System does not include any tertiary users. All identified stakeholders who directly interact with the platform are classified as either primary or secondary users based on their operational responsibilities and level of system interaction.

USER RESEARCH & PAIN POINT ANALYSIS

1. Research Objective

The objective of the user research was to understand the operational challenges faced by study libraries and identify the needs of different stakeholders, including administrators, library staff, and students. The research focused on existing workflows related to membership management, seat booking, attendance tracking, payments, and communication to determine opportunities for improving efficiency through a Study Library Management System (SLMS).

2. Key Observational Insights

The following observations were made after studying the existing workflow and interacting with stakeholders:

- Most study libraries rely on manual registers or spreadsheets to manage student memberships, attendance, and seat allocation.
- Staff spend significant time verifying memberships, assigning seats, and recording daily attendance.
- Students often visit the library without knowing whether seats are available, leading to unnecessary waiting.
- Booking modifications and cancellations are managed manually, making it difficult to utilize vacant seats effectively.
- Payment records are maintained separately, increasing the risk of errors and making transaction tracking difficult.
- Communication regarding announcements, membership expiry, and complaints is handled informally through phone calls or messaging applications.
- Administrators lack centralized reports on occupancy, memberships, revenue, and peak usage hours, making operational planning challenging.

3. Stakeholder Pain Points

3.1 Students

- Difficulty checking seat availability before visiting the library.
- No facility to reserve study seats in advance.
- Lack of visibility into membership validity and renewal dates.
- Difficulty accessing payment history and booking records.
- Limited communication regarding announcements, booking confirmations, and complaint resolution.
- Time wasted waiting for seat allocation during busy hours.

Core Issue: Students face difficulty accessing study library services efficiently due to limited visibility into seat availability, memberships, and bookings.

3.2 Library Staff

- Manual student registration and membership verification consume considerable time.
- Maintaining attendance registers is repetitive and prone to human error.
- Managing walk-in students alongside advance bookings is difficult.
- Tracking payments manually increases administrative workload.
- Handling complaints without a centralized tracking system delays resolution.
- Difficulty identifying no-show students and reallocating unused seats promptly.

Core Issue: Library staff spend excessive time performing repetitive manual tasks, reducing operational efficiency and increasing the likelihood of errors.

3.3 Admin

- Limited visibility into overall library performance.
- Difficulty monitoring occupancy trends and peak usage hours.
- Time-consuming management of memberships and membership plans.
- Lack of consolidated reports for revenue, attendance, and memberships.
- Inefficient communication with students regarding important announcements.
- Limited ability to configure and enforce booking policies consistently.

Core Issue: Administrators lack centralized control and real-time operational insights, making effective management and decision-making challenging.

4. Cross-Stakeholder Patterns

Several common issues were identified across all stakeholders:

- Heavy dependence on manual record-keeping.
- Duplicate data entry across multiple registers and spreadsheets.
- Delays caused by manual verification and approval processes.
- Lack of real-time visibility into seat availability and attendance.
- Difficulty accessing historical records quickly.
- Inefficient communication between administrators, staff, and students.
- Increased likelihood of human errors in attendance, bookings, and payment records.

5. Root Causes Identified

The primary causes contributing to the identified challenges include:

- Absence of a centralized digital platform for managing study library operations.
- Manual processes for bookings, attendance, payments, and memberships.
- Lack of real-time synchronization of seat availability.
- Limited automation for routine administrative tasks.
- Fragmented storage of operational data across multiple records.
- No integrated notification mechanism for students and staff.
- Limited reporting and analytics capabilities for decision-making.

Overall Conclusion

The user research indicates that the current management of study libraries is largely manual, resulting in inefficiencies, inconsistent records, and increased administrative effort. Students experience inconvenience due to limited visibility into seat availability, memberships, and bookings, while library staff spend considerable time on repetitive operational tasks. Administrators lack comprehensive insights needed for effective decision-making.

The proposed **Study Library Management System (SLMS)** addresses these challenges by providing centralized membership management, real-time seat booking, attendance tracking, payment management, complaint handling, automated notifications, and analytical reporting. By digitizing these processes, the system aims to improve operational efficiency, reduce manual errors, enhance resource utilization, and deliver a better overall experience for administrators, staff, and students.

USER STORIES

(The following user stories follow the S.M.A.R.T guidelines (Specific, Measurable, Achievable, Relevant, Time-bound) Format: As a [User], I want [Action], So that [Benefit].)

Epic 1: User & Membership Management

Story 1.1: Staff Account Management

- **As an** Admin,
- **I want** to create, update, and deactivate library staff accounts,
- **So that** only authorized staff members can access the system.

Story 1.2: Student Membership Management

- **As an** Admin,
- **I want** to approve and manage student memberships,
- **So that** only eligible students can use the study library.

Story 1.3: Student Registration

- **As a** Library Staff,
- I **want** to register new students and verify their memberships
- **So that** students can access the study library without delays.

Story 1.4: Student Account & Membership Information

- **As a** student,
- I **want** to securely log in to my account, reset my password if needed, and view my membership details and validity,
- **So that** I can securely access the system, manage my account, and know my membership status and renewal date

Story 1.5: Profile Management

- **As a** Student,
- **I want** to update my personal details,
- **So that** my contact information remains up-to-date

Epic 2: System Administration & Configuration

Story 2.1: Operating Hours Configuration

- **As an** Admin,
- **I want** to configure the operating hours of the study library
- **So that** students can only book and access the library during working hours.

Story 2.2: Booking Limit Configuration

- **As an** Admin,
- **I want** to set booking limits, including the number of available seats,
- **So tha**t occupancy is managed efficiently.

Story 2.3: Membership Plan Management

- **As an** Admin ,
- **I want** to create and manage membership plans,
- **So that** students can subscribe to the plan that best suits their needs.

Story 2.4: Library Reports & Analytics

- **As an** Admin ,
- **I want** to view reports and statistics on occupancy, peak hours, memberships, and revenue,
- **So that** I can make informed operational decisions.

Epic 3: Booking and Seat Management

Story 3.1: Real-Time Seat Availability

- **As a** Student,
- **I want** to view real-time seat and study room availability
- **So that** I can choose an available space.

Story 3.2: Advance Booking

- **As a** Student,
- **I want** to book study seats or rooms in advance
- **So that** I can reserve a study space before arriving

Story 3.3: Booking Modification

- **As a** Student,
- **I want** to modify or cancel my booking within the permitted time
- **So that** I can adjust my reservation if my plans change

Story 3.4: Booking and Attendance History

- **As a** Student,
- **I want** to view my booking and attendance history
- **So that** I can review my study sessions

Story 3.5: Daily Booking Overview

- **As a** Library Staff,
- **I want** to view daily bookings and expected arrivals
- **So that** I can prepare for incoming students

Story 3.6: Walk-In Admission

- **As a** Library Staff,
- **I want** handle same-day walk-in admissions by assigning available seats
- **So that** students without reservations can use the facility

Story 3.7: Extension Approval & Seat Allocation

- **As a** Library Staff,
- **I want** to review extension requests and either extend the current booking or assign a new available seat,
- **So that** students can continue their study sessions while ensuring efficient seat utilization.

Epic 4: Check-in & Attendance Management

Story 4.1: Student Check-In

- **As a** student,
- **I want** to present my unique check-in code to the library staff,
- **So that** my attendance can be verified before I access the study library.

Story 4.2: Check-In Verification

- **As a** Library Staff,
- **I want** to verify booking details using the student's check-in code,
- **So that** only students with valid bookings are checked in.

Story 4.3: No-Show Management

- **As a** Library Staff,
- **I want** to mark no-show bookings after waiting time and release reserved seats,
- **So that** unused seats become available to other students

Story 4.4: Study Session Check-Out

- **As a** Student,
- **I want** to check out after completing my study session,
- **So that** my attendance is recorded accurately and my seat becomes available for other students.

Epic 5: Payment Management

Story 5.1: Offline Payment Recording

- **As a** Library Staff,
- **I want** to record student payments and specify the payment mode, such as cash, UPI, or card,
- **So that** all transactions are documented accurately.

Story 5.2: Payment History

- **As a** Student,
- **I want** to view my payment history,
- **So that** I can keep track of all my library transactions.

Epic 6: Complaints & Communication

Story 6.1: Complaint Submission

- **As a** Student,
- **I want t**o raise complaints to the library staff,
- **So that** my issues can be addressed promptly.

Story 6.2: Complaint Management

- **As a** Library Staff,
- **I want** to view and resolve student complaints,
- **So that** reported issues are handled efficiently and in a timely manner.

Story 6.3: Announcements & Notifications

- **As an** Admin,
- **I want** to create and manage announcements and notifications
- **So that** important information can be communicated to all users.

Link of Interviews and Proofs of Interaction: - Interviews and Proofs